

FROM CENTRAL PLACE TO NETWORK MODEL: THEORY AND EVIDENCE OF A PARADIGM CHANGE

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ABSTRACT

While the deficiencies of the central place model have often been highlighted, no other paradigm has replaced it. However, recently some researchers have hinted at the development of a new model of spatial organisation, a network model. This model would hold most in polycentric urban regions. This paper discusses the features of this network model in comparison with the central place model. Moreover, it explores whether this model describes spatial reality better, thereby focusing on complementarity, a main feature of the model. The relationships within multi-location hospitals and universities of professional education (*hogescholen*), which spread their offer of care and study programmes over multiple, close-by cities, are analysed for this reason. Within the hospital care sector there is a clear trend towards complementarity, in line with the network model. The *hogescholen* sector provides a more ambiguous picture. The network model, however, still seems more appropriate than the central place model.

Key words: Central place theory, network model, complementarity, polycentric urban regions, urban facilities, correspondence analysis

INTERCITY RELATIONSHIPS: FROM HIERARCHY TO NETWORK?

The study of the spatial structure of urban systems, and intercity relationships in particular, has occupied an important place in the fields of urban planning, regional science, urban and economic geography for a long time. Notably, the work of Christaller (1933) and Lösch (1941) on what has come to be known as central place theory launched this strand of research. In the 1960s it became an important part of urban systems research. By defining sets of cities in terms of systems, explicit attention was drawn to the interdependency of cities and developments in other cities in a region or nation, and how such linkages affect growth and development. However, while this search for theories, rules and other

regularities that describe patterns in sets of cities flourished in the 1970s and early 1980s, interest for urban systems research has been on the wane ever since, and central place theory has disappeared from policy discourse (Coffey 1998).

One of the explanations for this is that the basic theoretical principles that are part of central place theory's conceptual core, such as the notion of hierarchy, have been challenged (Davies 1998). Though in the heyday of urban systems research many enhancements and refinements of the initial work by Christaller and Lösch were introduced by authors such as Berry, Dacey, Mulligan, Beckmann, Beguin, Parr and others (see Berry *et al.* 1988, and Coffey *et al.* 1998 for an overview), the central place model has had increasing difficulties explaining spatial reality. While the deficiencies of the central

place model have been highlighted often, no other set of clearly defined hypotheses has replaced those of central place theory (Camagni 1993).

However, from the early 1990s onward, some researchers have hinted at the development of a new model of spatial organisation, which is generally referred to as a 'network model' (see Camagni 1993; Batten 1995; Van der Knaap 2002). This model is essentially opposite to the central place model. Its basic principles can be dated back to the discussion on the concept of a 'dispersed city' (Burton 1963): a group of rather similar-sized politically discrete cities, separated by tracts of open land, functioning economically as a single urban unit. Not only is the central place model at odds with such a sudden clustering of relatively similar-sized cities, evidence suggested that such dispersed cities were also not arranged in an hierarchical pattern due to local specialisation. Instead of 'dispersed city', the now common label for such regions is 'polycentric or polynuclear urban region' (see Kloosterman & Musterd 2001; Davoudi 2003; Parr 2004), while also the terms 'polynucleated metropolitan regions' (Dieleman & Faludi 1998), 'city network' (Camagni & Salone 1993) and 'network city' (Batten 1995) have been used. It is precisely in such a spatial setting of a group of relatively similar-sized close-by cities that it is most evident that the central place model does not hold true (Camagni 1993; Davies 1998; Capello 2000; Van der Knaap 2002). Consequently, this network model would hold most for such polycentric urban regions. However, while the theoretical framework of the network model paradigm seems relatively well established, research demonstrating its empirical validity is largely non-existent (Capello 2000).

The aim of this paper is to describe the features of this network model in comparison with the central place model as well as to assess whether this network model describes the spatial pattern in polycentric urban regions better than the central place model, at least for some of its features. As central place theory, in particular Christaller's approach, gradually developed into a descriptive and theoretical base of the geography of retail and consumer services (Von Böventer 1963; Berry *et al.* 1988), we will focus in our elaboration on consumer-oriented services as well: hospitals and universities of professional

education in the Netherlands. Due to processes of scaling-up of supply and demand, hospitals and institutes for higher education have turned into regional rather than local facilities; which makes them an interesting case. Increased competition for patients/students and the introduction of 'the market' in both sectors has probably meant that their hinterland is becoming less connected to the definition of non-overlapping market areas. In addition, as a result of mergers, many multi-location institutions have developed in both sectors, spreading their activities over multiple cities in a region. According to Taylor (2003), who, similar to Pred's (1977) approach, argues that intercity relationships can be empirically studied and understood by looking at the relationships between the locations of multi-location organisations, the study will focus its attention on these multi-location organisations as well. Consequently, the research question is whether the spread of activities in the hospital and higher education sectors in polycentric regions obeys the rules of the central place model or whether it justifies the network model.

The second section describes the principal features of this network model in more detail. The third section presents the research approach. The analysis of the higher education sector (universities of professional education) is presented in the fourth section, the analysis of the hospital sector in the fifth section. The final section compares the developments in both sectors and explores whether these underline the rise of a network model of spatial organisation.

A NETWORK MODEL OF SPATIAL ORGANISATION

The ideas on a network model of spatial organisation have sprung up in response to the deficiencies of the central place model. While the focus here will be on the features of this network model, there will first be a brief discussion on the basics of the central place model.

Christaller posited that each commodity has a given threshold of minimum demand as well as a fixed geographical domain beyond which people are unwilling to pay for. Therefore, only a certain proportion of all settlements will offer higher-order goods and services. The variety of consumer goods and services offered by establishments in cities (or central places) of a

certain class, or order, and of a given size is dependent upon the number of thresholds the combined population of the city and its hinterland can fulfil. If these thresholds do not meet the minimum demand necessary for certain goods and services, then a central place must obtain that class of goods and services from the nearest more populous central place which does meet the threshold requirement in question. The largest central place in a country or region is completely self-reliant. Central places of each class are distributed evenly across the region. Central place theory puts great emphasis on one-sided vertical relationships between different classes of hierarchically-ordered central places. 'One-sided' means that the lower class of central places is dependent on the higher class of central places, not vice versa. Horizontal relationships between cities in the same class (thus of similar size) would be non-existent and also redundant, as these cities provide the same amenities and services. Although many enhancements and refinements have made the model less rigid (see Berry *et al.* 1988, for an overview), its essence has remained. However, nowadays it is generally acknowledged that real city systems in advanced economies have departed in many respects from the Christallerian pattern of a nested hierarchy of centres and markets.

While the discourse on alternative models to the central place model began to take shape in the early 1990s, this discourse was informed by some early observations of alternative spatial patterns, particularly in polycentric urban regions. Next to Burton's work on the dispersed city, Gottmann's work on the Megalopolis is a case in point (Gottmann 1961). In his analysis of the urbanised North-eastern Seaboard of the United States Gottmann opposed the dominant view of hierarchical relationships, and instead suggested that complementarity existed in this polycentric region. In addition, Pred (1977) analysed the spatial structure of multi-locational business organisations, arguing that hierarchies are linked to functions rather than cities. Hierarchies in functions are often not symmetrical to the hierarchy in cities, which implies that relationships between cities may not only be vertical but also of a horizontal nature. This also implies that complementarity between cities follows from hierarchies in functions that are asymmetric to the hierarchy in cities.

More recently, the spatial organisation of polycentric urban regions has been described by drawing analogies with economic linkages, or networks, among firms (Camagni & Salone 1993; Batten 1995), although this analogy is not undisputed, given the much stronger stability of urban systems over time (Pumain 1992). However, Camagni & Salone (1993) argue that if the shape of the urban hierarchy is determined by the interplay of forces such as economies of scale, minimum efficient production size, demand density and market size, as is the case in the Christallerian model, then it could well be that other production forces working at the micro-economic and micro-territorial scale of the firm may be considered as the driving forces of the new 'network' paradigm (Camagni & Salone 1993). Based on the resemblance with networks among firms, they define networks among cities as 'systems of relationships and flows, of a mainly horizontal and non-hierarchical nature, among specialised centres, providing externalities or economies respectively of specialisation/complementarity/spatial division of labour and of synergy/co-operation/innovation' (Camagni & Salone 1993, p. 1059). Such non-hierarchical relationships are also emphasised by Batten (1995, p. 313): 'A network city evolves when two or more previously independent cities, potentially complementary in function, strive to co-operate and achieve significant scope economies aided by fast and reliable corridors of transport and communications infrastructure'.

As both authors stress, the idea of horizontal and non-hierarchical relationships of complementarity and co-operation between cities in a polycentric urban region contrasts with the hierarchical, gravity-type relationships predicted by the central place model (see Table 1). Van der Knaap (2002) presents a comparable overview of the differences between both models (Table 2). However, rather than replacing the central place model with a network model, these authors suggest a sequential link between both models. Whereas the central place model seems most typical for industrial economies, the network model seems more applicable to economies that have become more service-sector dominated (Camagni & Salone 1993; Batten 1995; Van der Knaap 2002).

A number of the features attributed to the network model (as in Tables 1 and 2) have for a long time been accepted as better describing

Table 1. *Central place versus network systems.*

Central place system	Network system
Centrality	Nodality
Size dependency	Size neutrality
Tendency towards primacy and subservience	Tendency towards flexibility and complementarity
Homogeneous goods and services	Heterogeneous goods and services
Vertical accessibility	Horizontal accessibility
Mainly one-way flows	Two-way flows
Transport costs	Information costs
Perfect competition over space	Imperfect competition with price discrimination

Source: Batten (1995, p. 320).

Table 2. *Changes resulting from the transition from hierarchies to networks.*

Hierarchy	Networks
Fixed number of spatial scales	Variable number of spatial scales
Economic functions rising with spatial scale, functions connected to spatial scale	Variable sets of economic functions on the same spatial scale
Urban population evenly distributed across territory	Uneven distribution of urban population across territory
Only vertical relationships between cities (on different spatial scales)	Both horizontal and vertical relationships between cities

Source: Van der Knaap (2002, p. 168), translation by author.

spatial reality than the corresponding features of the central place model. This holds, for instance, for the idea of imperfect competition, a more flexible and variable number of spatial scales and the uneven distribution of urban population. A more innovative feature of the network model describing a non-hierarchical relationship between cities is the idea of 'complementarity'. This refers to the situation in which different cities fulfil different and mutually beneficial roles (Hague & Kirk 2003), for instance through providing different sets of economic functions and services. So, complementarity results from the differentiation between centres or cities in terms of urban functions, while these urban functions should be provided, at least partly, for the same geographical demand market (Ullman 1956; Meijers 2005). Complementarity can be considered a main feature of the network model as it positively enhances the presence of other characteristics of the network model. Complementarity results in two-way flows between both different and similar-sized cities, thus emphasising

also horizontal accessibility. Moreover, it may explain the aspect of size neutrality, which refers to a relative disconnection between size and function of a city. Higher-order functions may be found in smaller cities operating in a network because of complementarity, thereby drawing on the regional support base rather than the local. Consequently, complementarity is also linked to nodality, the position of a city in a network, rather than centrality.

Given these links between the features of the network model of spatial organisation, it seems that the feature of complementarity provides an excellent starting point for the comprehensive research agenda to test the empirical validity of the network model of spatial organisation. In the following sections the research question will be focused on whether the spread of activities in the hospital and higher education sectors in polycentric regions obeys the rules of the central place model or the network model on the feature of complementarity. Given the definition of complementarity above, the analysis should

focus on aspects of differentiation between locations of hogescholen and hospitals as well as on the origins of students and patients.

RESEARCH APPROACH

Hogescholen – The Netherlands has a binary system of higher education, which means that there are two types of programmes: research-oriented education (*wetenschappelijk onderwijs*, WO), traditionally offered by research universities and leading to bachelor, master's and Ph.D. degrees, and professional higher education (*hoger beroepsonderwijs*, HBO), traditionally offered by universities of professional education, or hogescholen (we will refer to them by this Dutch name), generally leading to bachelor's degrees. Research universities tend to operate on a supraregional or national level, less relevant to polycentric urban regions, while hogescholen function on a regional level, in theory more congruent with the scale of polycentric urban regions. Therefore, the analysis focuses on these hogescholen, using a dataset provided by the Informatie Beheer Groep in which all full-time study programmes offered by these hogescholen are registered (CROHO-database). The relationship between locations of a hogeschool are defined as complementary when they offer different study programmes while recruiting their students from more or less the same region. The dataset allows a quantitative analysis of the extent of differentiation in the offer of study programmes by using correspondence analysis techniques from which a complementarity ratio can be inferred. In addition, interviews were carried out with public sector officials and managers of large multi-location hogescholen to learn more about their locational strategies as well as their institutional context. The analysis was completed with a literature review.

Hospitals – In the Dutch hospital care system, general hospital care is provided by academic hospitals and general hospitals. Next to general care, academic hospitals, of which there are eight in the Netherlands, also deliver very specialised treatment for complicated disorders, for which they often have a 'last resort' function. The largest group of hospitals are the general hospitals, which provide a large variety of medical specialist care for their region. This study's analysis

focuses on the general hospitals as these provide the overwhelming majority of general hospital care, also functioning on a supralocal level, potentially that of a polycentric urban region, whereas academic hospitals function on a much higher and thus less relevant level. Locations of a multi-location hospital are considered as complementary when they provide for different medical specialisations, or specialise in different kinds of treatment while serving more or less the same region. No general datasets displaying aspects of complementarity are available in the Netherlands, which means that no quantitative analysis could be undertaken. Therefore, information was gathered during a series of interviews with managers of hospitals in the case study regions. In addition a literature review was conducted.

Multi-location organisations – Following Taylor (2003), who, similar to Pred's (1977) approach, argues that intercity relationships can be empirically studied and understood by looking at the relationships between the locations of multi-location organisations, the study will also focus its attention on multi-location organisations. By multi-location it is meant that these organisations have locations in multiple cities.

Study region – The multi-location model has become increasingly dominant in both the hospital and higher education sector. Both hospitals and hogescholen spread their activities over multiple cities. Consequently, these facilities now have regional significance and are potentially, though not necessarily, functioning on the scale of a polycentric urban region.¹ Such polycentric regions are spread all over the Netherlands, which may not be surprising given its high extent of urbanisation and high population density. In the Dutch spatial planning strategy polycentric urban regions are labelled 'urban networks' and the strategy distinguishes six of them being of national and international importance (see Van der Burg & Dieleman 2004). Though this study basically included all Dutch multi-location hogescholen for which data was available, these happen to all be located in these six urban networks distinguished in the national planning strategy. For the analyses of multi-location hospitals, hospitals were selected from two of these urban networks, namely the South

Wing of the Randstad and Brabantstad, for which the province of North Brabant is a good proxy. Camagni & Salone (1993) suggest that the Randstad is a prime example of a region in which complementary relationships have been developed.

HOGESCHOLEN

Macro-level context – Compared to neighbouring countries, institutions for higher education in the Netherlands are fairly autonomous (OECD 2004; Onderwijsraad 2005). This autonomy particularly concerns financial matters, but also spatially relevant decisions such as the closing, or opening of different locations of an institution in other places and co-operation or mergers with other institutions. In the past, Central Government has fostered a tremendous scaling-up in the hogescholen sector. In 1980 there were 353 hogescholen with an average size of 370 students, whereas in 2000 there were only 56 left with an average size of 4,460 students (Onderwijsraad 2004). According to the CROHO database, in 2005 there were 44 publicly-funded hogescholen remaining, offering full-time education at 114 locations (on average 2.5 locations per organisation). It follows that many hogescholen have multiple locations. In fact, this holds for 24 of these 44 hogescholen. This set of 44 hogescholen includes many locations offering just one full-time bachelor, which often happens to be teacher training for primary schools. Art degrees are also often given at very specialised locations or organisations. If these small unidimensional organisations and locations are excluded, that is, those offering just two or less full-time bachelor study programmes, then 35 organisations remain, spreading their offer of education over 67 locations. Nearly half of them have locations in a multitude of often close-by cities.

Although the Dutch Government has in general advocated deregulation since about 1985, it still accredits new and existing study programmes. It approves a proposal submitted by a hogeschool for a new study programme leading to an academic degree, irrespective of whether this concerns an entirely new study programme, the addition of a study programme at a particular hogeschool that is already given at another hogeschool, or, most relevant for multi-location

hogescholen, offering a study programme already given at one location also at another location of the same hogeschool (the latter since 2002). In the process of accreditation, Central Government is assisted by a special agency that tests whether the programme satisfies certain quality standards. If a study programme is also to be funded by the government, the Ministry of Education tests for what is called 'macro-efficiency' using criteria such as innovativeness, demand by employers for graduates and also the existing dispersal of similar study programmes across the country as they try to avoid harmful effects for the existing hogescholen. Whether or not they do test for macro-efficiency depends very much on the minister in office. The number of accredited study programmes remained limited to about 80 in the period from 1993 up to 1998. After that, with a more liberal Minister of Education (Hermans) in office, some 250 new study programmes were registered (Adviescommissie Onderwijsaanbod 2003). Recently, Central Government retraced their steps towards a more reserved approach, testing more rigorously for quality and macro-efficiency. Existing study programmes are nowadays also examined using criteria related to quality, but not of 'macro-efficiency'.

Trends in the differentiation in study programmes – The offer of study programmes of multi-location hogescholen was analysed using correspondence analysis. This is a method to analyse the association between two or more variables. These variables are the locations of multi-location hogescholen and the study programmes they offer. Correspondence analysis presents a statistic called the 'total inertia' expressing the extent of, in this case, the differentiation in study programmes between locations of multi-location hogescholen. The inertia is the weighted average of the χ^2 distances between the scores on both variables and the average scores. If all these locations provided exactly the same study programmes, the total inertia-statistic would be 0, thus representing maximum duplication. If all locations provided study programmes not given at any other location, then the total inertia would be equal to the dimensionality of the problem (in practice the number of locations of the organisation -1). For an extensive discussion of correspondence

analysis see Greenacre (1993). In order to enable comparisons between hogescholen that differ in terms of the number of locations, the total inertia statistic was normalised by dividing it by the maximum total inertia that could have been achieved. This then results in what is called a complementarity ratio, with values between 0 (no differentiation, thus maximum duplication) and 1 (maximum differentiation, and hence complementarity). For the analysis a dataset was used in which accredited full-time bachelor study programmes of hogescholen are listed. This CROHO registration was provided for the academic years 1993–1994, 1997–1998, 2001–2002 and 2005. The study selected all publicly-funded multi-location organisations and included all locations offering three or more full-time study programmes. However, data was not available in the database for each organisation, as it is only since 2005 that the database has included separate records for different locations rather than organisations. This meant that it was only possible to retrieve information for multi-location organisations resulting from a merger since 1993. The development in the differentiation in study programmes offered by the locations of these seven organisations is presented in Figure 1.

Figure 1 shows that there are huge varieties in scores on the complementarity ratio. The

study finds both multi-location hogescholen with a strong division of tasks between them, while others are characterised by strong duplication. In terms of trends, it seems that after the merger the locations duplicate rather than complement each other more. The average decrease in the extent of differentiation between 1993 and 2005 is 11.5 per cent. Though two hogescholen show an upward trend following a strong decline, none of the hogescholen scored above their 1993 and pre-merger value.

Of the seven multi-location hogescholen in Figure 1 the locations of the Hogeschool Zuyd complement each other most. Conversely, there is a high extent of duplication between the locations of the ArtEZ hogeschool and Avans hogeschool. Tables 3 and 4 provide a more detailed look at the Hogeschool Zuyd and Avans, as these are quite comparable regarding the number of locations (three) and the sectors in which they provide study programmes.

As Table 3 shows, the high score on complementarity for the Hogeschool Zuyd is because there is hardly any duplication of study programmes, even when its locations offer study programmes in the same sectors. This holds much less for the Avans hogeschool (Table 4), where each location offers study programmes that hardly makes them distinct from other

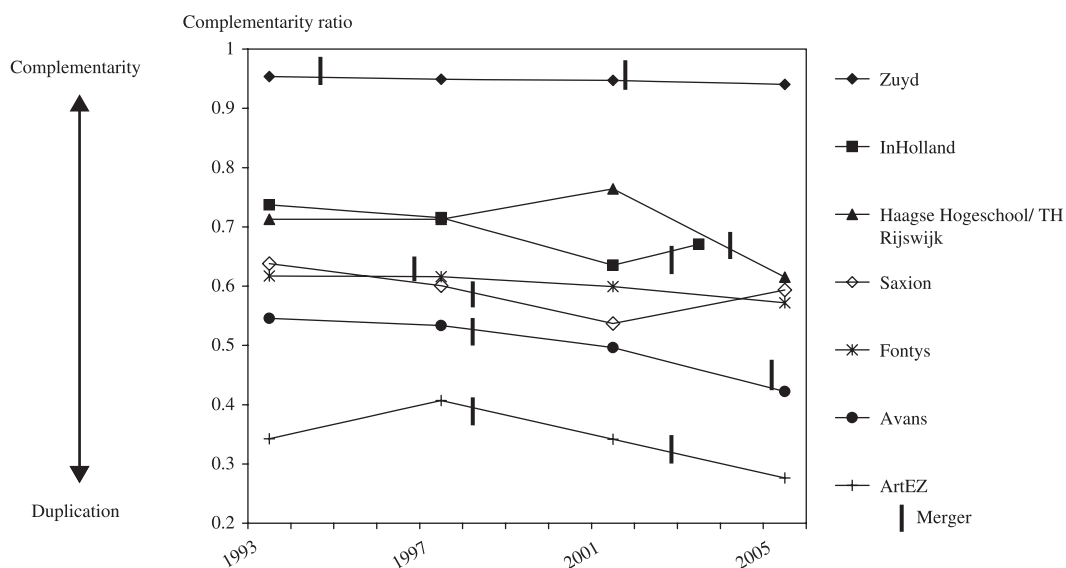


Figure 1. Trends in the extent of differentiation in study programmes within multi-location hogescholen, 1993–2005.

Table 3. *Hogeschool Zuyd: high extent of complementarity among locations (2005).*

Sectors	Total N of study programmes in sector	Location: Sittard-Geleen		Location: Heerlen		Location: Maastricht	
		Number of study programmes in sector	... that are not offered elsewhere	Number of study programmes in sector	... that are not offered elsewhere	Number of study programmes in sector	... that are not offered elsewhere
Economics	15	8	8	1	1	6	6
Social sciences	6	6	5	0	0	1	0
Health care	5	0	0	5	5	0	0
Education	4	0	0	1	0	4	3
Languages and arts	5	0	0	0	0	5	5
Technology	17	0	0	17	17	0	0
Total	52	14	13	24	23	16	14

Table 4. *Hogeschool Avans: limited extent of complementarity among locations (2005).*

Sectors	Total N of study programmes in sector	Location: Breda		Location: Tilburg		Location: Den Bosch	
		Number of study programmes in sector	... that are not offered elsewhere	Number of study programmes in sector	... that are not offered elsewhere	Number of study programmes in sector	... that are not offered elsewhere
Economics	13	8	2	4	1	10	3
Social sciences	4	4	0	0	0	4	0
Health care	3	2	2	1	0	1	0
Education	2	2	2	0	0	0	0
Languages and arts	2	2	0	0	0	0	0
Technology	18	11	5	8	2	11	2
Total	42	29	11	13	3	26	5

locations. Interestingly, however, it is not the case that the largest location offers all study programmes also offered by smaller locations, as we would expect in the case of a hierarchical central place model. How can these findings be explained?

Micro-level behaviour – Competition for students is an important factor in explaining the behaviour of the managers of hogescholen as the number of students successfully finishing their studies determines a hogeschool's share of the public budget for education. A certain amount

of competition has been eliminated through mergers, although there continues to be a couple of hogescholen competing in each region for students, so absolute monopolies are also avoided. An important aspect in this regard is the willingness of prospective students to travel for education. Some 53 per cent of the students of hogescholen prefer to stay at home with their parents rather than move to the city where they study (IB-groep 2005). One can imagine that this percentage is even higher for first-year students. Using a division of the Netherlands in 25 regions, Gordijn & Janssen (1997) found

that 80 per cent of the students of hogescholen studied in their own or an adjacent region in 1996. This means that, in general, students tend to choose the nearest hogeschool that offers their preferred study programme, even though Dutch students have free travel permits. For managers of such hogescholen it is not really possible to intervene in the spread of study programmes over their locations, as relocating study programmes from one place to another often means a loss of students. So, even after mergers, study programmes that are similar but given in different locations are not concentrated for the sake of scale economies. On the contrary, they often try to expand the offer of individual locations with study programmes already given at other locations of the same hogeschool. This is a relatively inexpensive way to attract more students as it does not involve development costs, though one can argue that this may lower the number of students of the same programme at the other locations. This also holds for the development of new study programmes that often, despite their fashionable labelling, do not differ much from existing study programmes. Conversely, when a truly new study programme is developed and appears successful in attracting students, other hogescholen tend to copy it soon afterwards (Adviescommissie Onderwijsaanbod 2003). It is clear, therefore, that competition between the hogescholen leads to duplication rather than complementarity. With complementarity being such an important feature of the network model of spatial organisation, this does not match the network model.

HOSPITALS

Macro-level context – Traditionally, the role of government in hospital care is to facilitate and to set conditions for the hospital-care sector but the hospitals themselves are privately-run organisations. These conditions relate to the quality and accessibility of hospital care and in particular to the control of the costs of hospital care. The latter was the main rationale for a policy to increase the scale of the hospitals and simultaneously decrease the overall capacity, in terms of the number of beds, in the past decades. The scaling-up, coupled with concentration, led to a diminishing of the number of hospitals from 221 in 1954 to 197 in 1970 (Boot 1998); 86 were still

left in 2005. Moreover, these hospitals have scaled-up considerably in terms of the number of beds provided: from around 330 in 1980 to over 500 in 2002. The process of scaling-up seems to have come to a halt recently now that, for reasons of accessibility of hospital care, the role of small hospitals received renewed recognition and their funding was increased. In addition, from the viewpoint of introducing more free-market competition in the hospital care sector, a further merging and concentration of hospital care is unwanted, as this may lead to regional monopolies (NMA 2004). Government involvement has, however, since the 1990s been more reserved, and more issues are now left to the market.

Decisions relating to the spatial dispersal of regular hospital care, and hence whether or not complementarity develops, have been taken by the hospitals themselves for a long time. It was private incentives that determined the spatial dispersal of hospitals in the first place, and this is still practically the same today as the hospital institutions themselves decide to close down hospital locations, engage in co-operative partnerships or mergers, concentrate or deconcentrate hospital functions, or relocate hospital functions over several locations of the same organisation (College Bouw Ziekenhuisvoorzieningen 2000).

Micro-level behaviour – The behaviour of hospital managers can be explained by various rationales, including the maximising of quality in health care, the realisation of a certain volume in health care as well as maximising the turnover (Den Hartog 2004). In addition, Boot (1998) explains their behaviour as a pursuit of maximising the number of medical specialist services provided by the hospitals. In the Netherlands there are about 30 acknowledged medical specialist services and providing them all would mean that a hospital has reached its maximum possible size, as the number of beds they are allowed to have is linked to the medical specialist services provided. To realise this growth, the hospital needs a larger hinterland, for which they compete with other hospitals. Hinterlands widen through mergers, which is one of the explanations for the many mergers that took place in the sector. The relevance of all these mergers is that many multi-location hospitals were created. While in 1990 only 41 of the 128 general hospital organisations had more than one location

(Nienoord-Buré & Damen 1992), the 86 hospital organisations that existed in 2005 had spread their activities over 168 locations. Nowadays, more than half of all the organisations have two or more locations, though some of them do not provide clinical care, but ambulatory care during office hours (so-called 'buitenpoli'). If we leave these 'buitenpoli' aside, we still find 24 hospital organisations with two locations, seven with three locations, while two hospital organisations have even spread their activities over four locations (see Figure 2).

Grosso modo, multi-location hospital organisations have a choice between three future location models: (i) having two or more locations offering the same care and treatment; (ii) concentrating their activities in one location; and (iii) relocating the care and treatment functions, resulting in

locations developing a functionally specialised profile, so that they complement one another. Options (i) and (ii) obviously conform to central place theories, while option (iii) would be a clear manifestation of a network model.

Which location model an organisation chooses depends on a number of factors and hence, is not necessarily the ideal model envisaged beforehand. These factors include the physical possibilities of the current locations: is there room for expansion? Financial matters play an important role as well. Developing a new building is costly, while often the existing locations have not depreciated fully in economic and technical terms. Also, government regulations with respect to building permits sometimes make the development of a new building (often involving a concentration of functions) impossible. One can

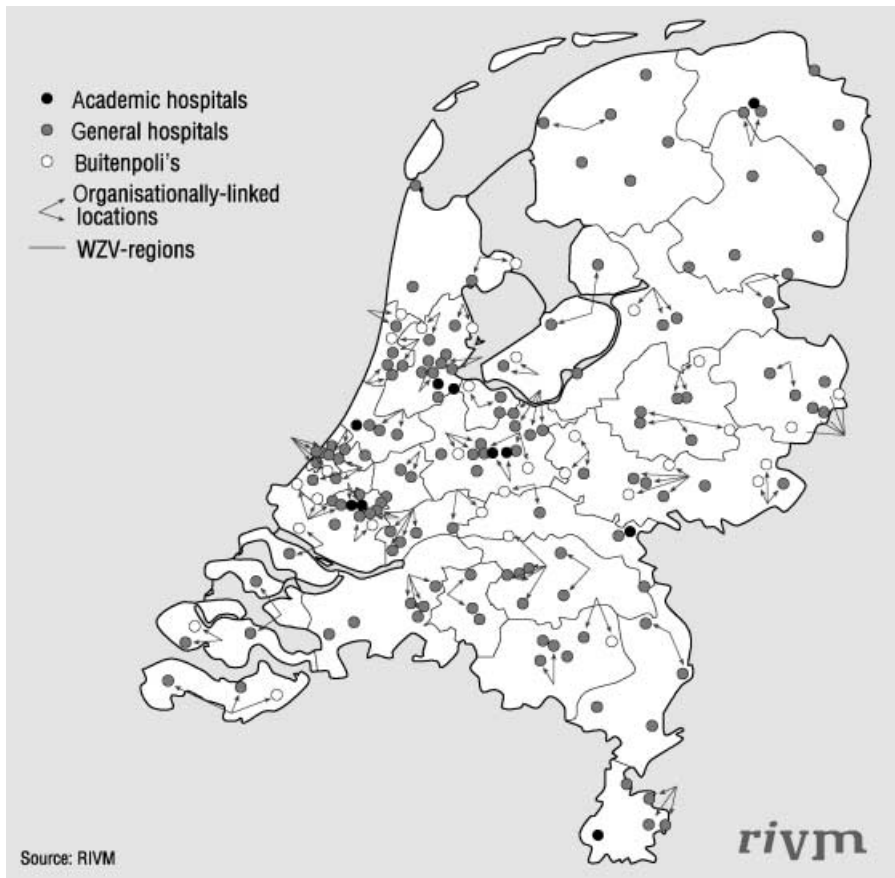


Figure 2. Hospital locations in the Netherlands, September 2004.

also imagine that for reasons of efficient operational management, concentration may be the most favourable situation, for instance because the facilities for laboratories and diagnostics, radiology, etc. can be used more efficiently. However, it may also prove to be very efficient to concentrate for instance, planned care in one location, so that specialists and staff become more practised and specialised and hence more people can be treated in a shorter period of time. Furthermore, the opinion of the medical specialists should not be overlooked. For them, having multiple locations and in particular specialised locations brings certain opportunities such as more tailor-made treatment for specific groups of patients. However, profiling locations also brings some practical difficulties, for instance, the necessity to travel between the locations. Finally, competition with other hospital organisations plays a major role. If closing down one location means a loss of hinterland to another hospital organisation, then closing down is not seriously considered. In the situation where the organisation has more or less a monopoly in the region, concentration in one location may be more feasible.

The outcome of the balancing of all these factors is increasingly often the specialisation of different locations in certain hospital care functions. This is particularly true in strongly urbanised city-regions, where multiple hospital organisations coexist close to each other and competition for the hinterland is fierce, so closing down locations is becoming uncommon. In these regions, for instance in the Randstad, we even see the opening of a number of *buitenpoli*, which should be considered as outposts safeguarding or extending the hospital's hinterland. When competition is fierce and the multi-location model is seen as a necessity, a hospital's board of directors often decides to give distinct, complementary profiles to the locations so that the organisation benefits from the potential advantages of the merger. Table 5 summarises the study's findings from interviews, strategic policy documents and hospitals' annual reports for all multi-location hospitals in the South Wing of the Randstad and the province of North-Brabant that resulted from a merger since 1990. No multi-location hospitals developed before that year, nor have any of the mergers in these regions since 1990 resulted in a concentration.

Table 5 shows that the majority of multi-location hospitals in both polycentric urban regions is implementing a functional specialisation of locations based on the type of care a patient needs: often one location specialising in emergency and complex care, another one in planned or elective care and sometimes we find locations specialised in high-frequency care for chronic patients or focusing on women or children. The extent to which they have implemented their desired location model varies. In general they will continue to offer planned care also at the location for complex and emergency care, however only for patients with a high risk for complications. Three out of the 10 multi-location hospitals will concentrate their activities on one location in the near future. One of them will continue to have two locations, but one will be for experimentation offering care on a private basis.

Given the definition, for hospitals to be complementary they should not only provide for different specialisations, but they should also serve more or less the same region. In the Netherlands, the hinterland of a hospital is termed its 'clinical adherency', that is the number of potential patients in the surrounding region, which is based on the share the hospital had previously in each municipality. Practically all mergers take place between hospital care organisations located in municipalities in which already a certain share of the population visits the other hospital. This means that the condition for complementarity, that the demand should come from more or less the same region, is met. As a result of profiling, for some types of hospital care, patients can no longer get help from the nearest hospital location. It appears that, in general, there is a slight fall in the number of patients following the profiling of locations, but this tends to be adjusted within a couple of years. Moreover, it seems to be more due to a lack of clarity than to an unwillingness of some patients to travel further. A recent survey (Ecorys-NEI 2003) showed that, alongside the expertise of the medical staff, travel time is the most determining factor in the patient's choice of a hospital. Also, some 40 per cent of the respondents would prefer to get treatment in a considerably more distant hospital if they were treated much sooner than in the nearest hospital. Of course, for elective, planned care the willingness to travel is much larger than for

Table 5. *Multi-location hospitals in the South Wing and Brabantstad and their desired future location model.*

Multi-location hospital organisation	Year of merger	Locations	Desired location model	Implementation
Rijnland Ziekenhuis	1990	Leiderdorp; Alphen aan den Rijn	Multi-location, locations having focused profiles (Leiderdorp: C; Alphen aan den Rijn: P)	Advanced
Reinier de Graaf Groep	1996	Delft; Voorburg	Multi-location, locations having focused profiles (Delft: C; Voorburg: P)	Advanced
TweeSteden Ziekenhuis	1997	Tilburg; Waalwijk	Multi-location, locations having focused profiles (Tilburg: C; Waalwijk: P)	Advanced
Medisch Centrum Haaglanden	1998	The Hague; Leidschendam	Multi-location, locations having focused profiles (The Hague: C; Leidschendam: P)	Start/plan phase
Vlietland Ziekenhuis	1999	Schiedam; Vlaardingen	Concentration	Concentration from 2007
Albert Schweitzerziekenhuis	1999	Dordrecht (2); Zwijndrecht; Sliedrecht	Multi-location, locations having focused profiles (Dordrecht 1: C; Dordrecht 2: P; Zwijndrecht: C; Sliedrecht: P)	Advanced
Ziekenhuis Bernhoven	2000	Veghel; Oss	Concentration	Start/plan phase
Amphia Ziekenhuis	2001	Breda (2); Oosterhout	Multi-location, locations having focused profiles (Breda 1: W+CH+CHR; Breda 2: C; Oosterhout: P)	Intermediate
Jeroen Bosch Ziekenhuis	2002	Den Bosch (3); Boxtel	Concentration (Den Bosch: C; Boxtel: PR)	Concentration from 2009
Máxima Medisch Centrum	2002	Eindhoven; Veldhoven	Multi-location, relative similar profiles	Continuation of pre-merger situation

Notes: C = Complex and emergency care, CH = Children, CHR = Chronic patients, P = Planned, elective care, PR = Privately funded care, W = Women.

emergency or chronic care. Moreover, it appears that in general young people are more willing to travel than older people, who account for the largest demand.

CONCLUSION

According to Kuhn (1962), a crisis over the tenability of a paradigm may end with the emergence of a new candidate for a paradigm and with the ensuing battle over its acceptance. This paper explored whether the network model of spatial organisation could be such a new paradigm replacing the criticised central place model, at least in polycentric urban regions. An important feature of this network model is that functions such as urban facilities are spread over the different cities in such a way that they complement each other. This paper explored the development of this complementarity, thereby focusing on two urban facilities, hospitals and hogescholen, as these are increasingly organised on a regional level, which takes shape through the development of multi-location organisations. They spread their locations over a number of close-by cities, more or less comparable to the notion of a polycentric urban region. Therefore, relationships between the locations of these multi-location organisations may be considered exemplary for intercity-relationships.

Can these relationships better be described by central place theory or is the network model more appropriate? Empirical evidence for the hospital sector supports the latter. A large number of the multi-location hospitals choose to give a more distinct profile to their locations leading to complementarity. The old situation of each city having its own hospital providing as many specialist services as allowed by the size of their hinterland was very much in line with the central place model. However, the development of institutional ties between hospitals brings along a trend that hospital managers seek scale economies through imposing a division of labour. Many of these multi-location hospitals are therefore characterised by a high extent of complementarity, thus supporting the network model. This specialisation is also forced by a strong need for efficiency given the high costs of hospitals and care.

Although the macro-context for the hogescholen sector is quite similar, the actual micro-level behaviour of managers of multi-location hogescholen diverges from those of hospitals. Multi-location hogescholen tend to copy study programmes rather than profiling their locations, a trend that contradicts the network model. Part of the explanation is that the costs of duplication are small for hogescholen, while many prospective students are also reluctant to travel far. On the other hand, it was possible that even in a situation with few complementarities between locations, the largest location did not offer all the study programmes offered by smaller locations, as one would expect in the case of a hierarchical central place model. This supports the idea of size neutrality, another feature of the network model. Even while the analysis of the hogescholen sector does not unambiguously support the network model of spatial organisation, as does the hospital sector, the findings do not support the central place model either.

While this exploration was necessarily limited in terms of scope, its results emphasise the necessity and value of further exploration of the comprehensive research agenda of whether a network model of spatial organisation is developing. Urban networks research could then become the contemporary pendant of what urban systems research used to be.

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Note

1. Note that polycentric urban regions as identified in the literature and by policy-makers differ substantially in scale, from large metropolitan regions as the Randstad or the RheinRuhr Area to a clustering of small cities and towns. Here, we argue that these supralocal facilities potentially operate on a scale that includes at least two cities of a polycentric urban region.

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